Charlie Chavez Meelboom

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Education

San Diego State University — **San Diego, CA**
B.S. Mechanical Engineering | Minor: Mathematics | GPA: 3.81 — *Expected May 2028*
Coursework: Mechanics of Materials, Fluid Mechanics, MATLAB, Physics 3 (Thermodynamics & Optics), Mechatronics

Experience

Aztec Electric Racing (Formula SAE Electric) — **San Diego, CA**
Powertrain Engineer — *Sep 2024 – Present*

- Owned mechanical design of safety-critical housings (RML, Fill Can) in SolidWorks; iterated Bambu X1 Carbon prototypes to validate fit, assembly, and sealing, delivering a fully waterproof RML housing.
- Supported design through Reset Latch PCB documentation; GD&T component powertrain drawings, test-data collection for design validation, and powertrain tab-thickness structural analysis.
- Supported manufacturing through rear-wing carbon-fiber wet layups, waterjet fabrication for tabs and rotors and 3D printing of housings.

SDSU Dynamic Systems and Intelligent Machines (DSIM) Lab — **San Diego, CA**
Undergraduate Researcher — *Aug 2025 – Present*

- Support AUV acoustic-localization project by documenting pool-test procedures to improve repeatability and data quality. Designed hydrophone mounts and tight-tolerance housings with clean cable routing, and integrated the full sensor/servo stack.
- Developed the data pipeline using ROS2 nodes and Arduino IDE that relays sensor angle outputs from Rasberry Pi and DAC HAT for processing within an extremum seeking control framework. Implementing localization and obstacle-avoidance sensing.

KDG Industrial — **San Diego, CA**
Sales & Warehouse Clerk (Seasonal) — *Sep 2022 – Jan 2025*

- Reduced quoting times by $\sim$ 3 days for a high-volume international customer account through tighter coordination with technical and warehouse teams and weekly task meetings with team.
- Implemented HubSpot CRM workflows and training documentation that became the company's primary deal-flow platform; oversaw material reception, inventory checks, and local deliveries, and supported product exports to Mexico.

Projects

Self-Sorting Waste Bin Prototype (Innovate 4 SDSU Hackathon)
3rd Place ("Most Impactful")

- Built a live demo that classifies compost/landfill/recycling from real-time sensor readings and actuates sorting behavior using an Arduino-based sensor setup. Developed a KNN classifier achieving 74% accuracy; led the final pitch and demo.

Robotic Arm
Personal Project

- Designed a 6-DoF robotic arm in SolidWorks; 3D-printed components and implemented Arduino-based servo control with Python remote operation.

Leadership

Aztec Electric Racing (Formula SAE Electric) — **San Diego, CA**
Operations Lead — *Jun 2025 – Present*

- Built AER's first business/operations department to address a complete organizational gap; recruited 20 members, ran weekly execution meetings, and helped secure and over $120,000 in funding through alumni events and active sponsorship outreach.
- Led cost-report effort twice,leading sub team coordination of GD&T drawing collection and revisions to deliver 700+ page FSAE compliant report in under four months.

SDSU College of Engineering Student Council — **San Diego, CA**
VP of Finance (Elected) — *Jun 2025 – Present*

- Built a budgeting and tracking system and rewrote reimbursement documentation into a clear, repeatable process used by all engineering organizations at SDSU. Increased funding from university by $7,000 by lobbying with head of financial affairs.

Skills

Proficient in Arduino IDE, Python, Git, 3D Printing, Soldering/Wiring, Waterjetting, Test Documentation, ANSYS Mechanical, MATLAB, ROS2, Linux
Languages: English (fluent), Spanish (fluent)